Addition of New Course(s)

Approval and Inclusion of "Introduction to Quantum Computing" Course in BSCS, BESE, BSAI, BSDS Curriculum as Elective

SEECS,NUST

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Background

1. Quantum computing is a computer science discipline based on the principles of quantum mechanics. This state-of-the-art technology uses quantum mechanics to solve more complex problems than standard computers. As government and private sector entities increasingly prioritize quantum research and development, the quantum computing market is expected to expand significantly, reaching an estimated $6,528.8 million by 2030.

This working paper aims to include the course "Introduction to Quantum Computing" (with 3+1 and 2+1 variants) as an elective in the BESE, BSCS, BSAI, and BSDS degree programs. With Quantum Computing (QC) emerging as a transformative technology, this course is imperative to equip students with skills for future cryptography, optimization, and data science advancements. The course will provide foundational knowledge and practical skills essential for addressing complex problems using quantum paradigms.

Rationale

2. The inclusion of "Introduction of Quantum Computing" as an elective in BESE, BSCS, BSAI, and BSDS is justified for several reasons:

a. *Emerging Field:* With its potential to revolutionize fields like data science, cryptology, and telecommunications, quantum computing is rapidly emerging as a key driver of technological innovation. By adding a QC course, the BESE, BSCS, BSAI and BSDS programs can ensure that students are exposed to this cutting-edge field and prepared to meet the demands of the industry.

b. *Job Demand:* According to McKinsey, the QC job market faces a significant talent gap. There is a high demand for skilled quantum computing professionals, but the supply of qualified candidates is limited. By the end of 2025, it is predicted that less than 50% of quantum computing jobs will be filled unless significant steps are taken to address the talent shortage. This presents a unique opportunity for those with the right skills and expertise to secure high-demand, high-paying positions in a rapidly growing field.

c. Interdisciplinary Collaboration: Quantum computing demands interdisciplinary collaboration. Computer scientists and software engineers can collaborate with mathematicians for algorithm development, electrical engineers for hardware design, data scientists for machine learning, and cryptographers for quantum security. Such collaborations accelerate innovation and solve complex problems.

d. *Enhancing Creativity and Innovation*: QC provides a unique platform for creative problem-solving and innovation. By exposing students to QC techniques and algorithms, the degree programs can stimulate their creativity and inspire them to develop novel applications and solutions, fostering an entrepreneurial mindset.

e. *Hands-on Experience*: Incorporating a QC course will offer students the opportunity to gain hands-on experience in designing and implementing QC algorithms and systems. Practical assignments and projects can allow students to work with QC libraries and frameworks, enabling them to understand quantum algorithm design's technical challenges and intricacies.

f. *Future-proofing Graduates:* Quantum computing is rapidly emerging as a transformative technology. By incorporating a quantum computing course, BESE, BSCS, BSAI and BSDS programs can equip graduates with the skills needed to contribute to advancing this cutting-edge field.

3. In conclusion, the inclusion of an Introduction to Quantum Computing course in the BESE, BSCS, BSAI, and BSDS curriculum is justified due to its relevance in the job market, the interdisciplinary nature of this field, its potential for creativity and innovation,

and the need to equip students with practical experience in this rapidly growing field. These factors suggest that both degree programs should offer this course. By doing so, they can provide students with a well-rounded education and enhance their prospects in the evolving technology landscape.

Educational Objectives

4. This course provides a comprehensive introduction to quantum computing. It covers foundational concepts such as superposition, entanglement, and teleportation. Simple quantum algorithms and their complexity analysis will be discussed to compare quantum computing with classical computing. The course also explores emerging trends, emphasizing the applications of technology in various fields, such as cryptography and information retrieval.

5. The primary educational objectives of the course are:

a. Comprehensive understanding of quantum principles and algorithms.

b. Practical application through open-source tools like IBM Qiskit.

c. Development of problem-solving skills with quantum use cases.

d. Appreciate the interdisciplinary nature of Quantum Computing by exploring its connections with fields such as mathematics, computer science, cryptology, and data science.

e. Ability to critically evaluate existing QC algorithms and approaches and propose improvements and innovations.

f. Opportunity to collaborate with peers on projects, fostering teamwork and communication skills essential for working in interdisciplinary environments.

g. Inspiration for the students to engage in research and innovation in QC, encouraging them to explore new ideas, algorithms, and applications.

6. By achieving these educational aims and goals, students should be well prepared to pursue careers in telecommunications, cryptology, data science, and artificial intelligence, where QC expertise is increasingly in demand.

United Nations SDG Compliance

7. The "Introduction to Quantum Computing" course aligns with several United Nations Sustainable Development Goals (SDGs) by empowering students to apply their knowledge and skills to address global challenges.

a. SDG 4: Quality Education: The QC course enhances the quality of education by providing students with practical skills and knowledge in a cutting-edge field, preparing them to tackle real-world problems and contribute to sustainable development initiatives.
b. SDG 9: Industry, Innovation, and Infrastructure: QC techniques drive innovation in various industries, including telecommunication, cryptology, data science, and artificial intelligence. Students learn to develop intelligent systems and optimize processes, leading to more efficient and sustainable infrastructure.

Input Obtained from Industry/Corporate Sector/Subject Specialists/Academia

8. Input obtained/received from the maximum of the following:

a. Industry Advisory Board (IAB)
   i. Uroosa Kiran

      Manager (Cyber Security)

      Horizon Tech Service, Islamabad, Pakistan

      uroosa@horizon.com.pk

      Integrating quantum computing into a Bachelor's degree can be seen as a progressive step for contemporary computing education. Though quantum computing is still in its exploration phase, it is expected to change the landscape of cryptographic systems, optimization, AI, and material science. By blending quantum mechanics with computer science/software engineering, students can develop a unique skill set that will be in high demand as quantum computing becomes more practical and accessible.

In my opinion, the quantum computing course should start with introductory material and grow into more specialized topics with regular reviews for the inclusion of topics as per technological advancements. Relevant topics and their inclusion in a 4-year degree program are proposed below: -

- Foundation Courses (1-3 semesters):
- Mathematical Foundation (Linear Algebra for quantum state vectors, matrix representations of quantum operations, Probability theory relevant to quantum measurement and uncertainty),
- Basic Quantum Mechanics (wave functions, Schrodinger equation, superposition, quantum entanglement)
- Basic Quantum Computing concepts (gates) and algorithms (Dijkstra, Shor, Grover, etc)
- Python for Quantum Programming (implementation of quantum gates)
- Intermediate Courses (4-5 semesters):
- Python for quantum programming (Advanced level)
- Practical labs for programming quantum algorithms with Qiskit, Cirq, or Q# using quantum computing simulators e.g., IBM Quantum Experience, Microsoft Azure Quantum, Google's Quantum AI, etc)
- Construction and Analysis of Quantum Circuits
- Advanced Courses (6-8 semester):
- Quantum Machine Learning
- FYP / Specialized Research (Applications of quantum circuits and/or algorithms in optimization, AI, cryptography, etc.)

ii. Kashif Rahim

Project Director

Common Criteria Pakistan

kashif@commoncriteria.gov.pk

Quantum computing is one of this era's most prominent emerging technologies, where classical hardware seems to have touched its limitations. Therefore, our graduate and undergraduate-level students must become familiar with this new era of computing. A course that focuses on key concepts of quantum computation and its implications in communication, optimization, security, etc. will be very useful in this regard. The course may be designed with the following core objectives:-

- Understand the principles of quantum mechanics and their application in quantum computing
- Explore quantum algorithms and their advantages over classical algorithms.
- Learn the basics of quantum hardware and quantum programming languages
- Develop problem-solving skills for quantum computing applications

b. Subject specialists from local universities
Focal person: Muhammad Hassan Masood

MCS, NUST

mcs@nust.edu.pk

This elective course is typically offered in the 5th semester, by which point students have already covered complex numbers, probability, and linear algebra. Hence, dedicating the initial weeks to these topics is not required. Restructure to include 2-3 weeks on quantum mechanics fundamentals. Visualization tools, the no-cloning theorem, and the block sphere before algorithms, ensuring depth precedes breadth for meaningful mastery.

c. Subject specialists from foreign universities

Dr. Umar Mujahid

Associate Professor, Department of Information Technology

Georgia Gwinnett College, USA

ukhokhar@ggc.edu

I strongly endorse the inclusion of "Introduction to Quantum Computing" in the undergraduate curriculum. With quantum threats on the horizon, especially in cybersecurity, this course will equip students with critical insights into post-quantum cryptography and quantum-resilient system design. It is a timely and necessary addition to our academic offerings.

9. Furthermore, feedback from IAB is appended as Annex A.

International Practices

10. Many highly ranked universities are offering Quantum Computing courses.

a. Massachusetts Institute of Technology (MIT)

1. QOS World Ranking: Ranked 1st (2024)

2. Course Page: MIT xPRO | Introduction to Quantum Computing

b. Stanford University

1. QOS World Ranking: Ranked 5th (2024)

2. Course Page: Introduction to Quantum Computing | Stanford Online High School

c. The University of Edinburgh, UK

1. QOS World Ranking: Ranked 22nd (2024)

2. Course Page: Course Catalogue - Course Catalogue - Introduction to Quantum Computing (INFR11099)

Reference Curriculum of International Disciplinary Bodies (if applicable)

11. N/A

Summary/Key points from Student Course feedback forms

12. N/A

Summary/Key points from Faculty Course Review forms

13. N/A

Proposed Timeframe of Commencement

14. Spring 2026.

Course Contents

10. Give details of the course on the following lines:

Two variants of the course, i.e., 3+1 and 2+1, have been proposed following the recommendation of DBS. The following are the details of the course.

Introduction to Quantum Computing (3+1)

a. Course Code CS-XXX

b. Title: Introduction to Quantum Computing

c. Credit Hours 3+1

d. Objectives: This course provides a comprehensive introduction to quantum computing. It covers foundational concepts such as superposition, entanglement and teleportation. To compare quantum computing with classical computing, simple quantum algorithms will be discussed along with their complexity analysis. The course also explores the emerging trends, emphasizing the applications of technology in various fields i.e., cryptography and information retrieval. The successful completion of this course will enable the students to familiarize with the core concepts of quantum computing.

e. Outcomes: At the end of the course, the student will be able to:

1. Describe the basics of spin, superposition, and entanglement.

2. Implement quantum circuits for the modelling of classical problems in the quantum realm.

3. Analyze quantum computing algorithms, including their design, functionality, and efficiency.

f. Contents with suggested contact hours

Week 1	Mathematical Foundations of Quantum Computing
Week 2	Quantum mechanics fundamentals and visualization tools overview
Week 3	Qubits and Non-Cloning Theorem
Week 4	Bloch sphere representation of qubits and the framework of quantum mechanics
Week 5	Quantum model of computation
Week 6	Quantum Circuit Design
Week 7	Super dense coding
Week 8	Teleportation
Week 9	**Mid-Semester Break**
Week 10	Quantum simulations of classical circuits
Week 11	Binary, Phase kickback and Phase Oracles
Week 12	Deutsch algorithm
Week 13	Deutsch-Jozsa algorithm
Week 14	Simon's algorithm
Week 15	Grover's search algorithm
Week 16	Project presentations
Week 17	Guest Lecture
Week 18	**End Semester Break**

g. Details of lab work, and workshop practice (if applicable).

Lab 1	Virtual quantum lab-Spin experiment
Lab 2	Virtual quantum lab- Laser beam experiment
Lab 3	Mach-Zehnder interferometer
Lab 4	Introduction to quantum simulator
Lab 5	Construction of Bell states
Lab 6	Quantum simulations of classical circuits
Lab 7	Binary Oracles
Lab 8	Phase Oracles
Week 9	**Mid-Semester Break**
Lab 10	Quantum teleportation
Lab 11	Deutsch problem

Lab 12	Deutsch-Jozsa problem
Lab 13	Simon's algorithm
Lab 14	Grover's search problem
Lab 15	BB84 algorithm
Lab 16	Open Ended Lab
Lab 17	Presentation/Demo
Week 18	**End Semester Break**

h. Recommended Reading (including Textbooks and Reference books with dates).

Textbook:

1. Hughes, Ciaran, et al. Quantum computing for the quantum curious. Springer Nature, 2021.
2. Yanofsky, Noson S., and Mirco A. Mannucci. Quantum computing for computer scientists. Cambridge University Press, 2008.
3. Kaye, Phillip, Raymond Laflamme, and Michele Mosca. An introduction to quantum computing. OUP Oxford, 2006.

Reference Material:

1. IBM Quantum platform courses
2. IBM Quantum platform tutorial

i. Assessments

- Quizzes
- Assignments
- MSE
- Project (Optional)
- ESE

Introduction to Quantum Computing (2+1)

a. Course Code CS-XXX

b. Title: Introduction to Quantum Computing

c. Credit Hours 2+1

d. Objectives: This course provides a comprehensive introduction to quantum computing. It covers foundational concepts such as superposition and entanglement. To compare quantum computing with classical computing, simple quantum algorithms will be discussed along with their complexity analysis.

e. Outcomes: At the end of the course, the student will be able to:

1. Describe the basics of spin, superposition, and entanglement.

2. Implement quantum circuits to model classical problems in the quantum domain.

3. Analyze quantum computing algorithms.

f. Contents with suggested contact hours

Week 1	Mathematical Foundations of Quantum Computing
Week 2	Quantum mechanics fundamentals and visualization tools overview
Week 3	Qubits and Non-Cloning Theorem
Week 4	Bloch sphere representation of qubits and the framework of quantum mechanics
Week 5	Unitary quantum gates
Week 6	Quantum model of computation
Week 7	Quantum Circuit Design
Week 8	Binary Oracles
Week 9	**Mid-Semester Break**
Week 10	Deutsch problem
Week 11	Deutsch-Jozsa problem
Week 12	Function analysis through the Deutsch-Jozsa algorithm
Week 13	Quantum teleportation
Week 14	Grover’s search problem-Implementation
Week 15	Grover’s search problem analysis
Week 16	Project presentations
Week 17	Guest Lecture
Week 18	**End Semester Break**

g. Details of lab work, and workshop practice (if applicable).

Lab 1	Virtual quantum lab-Spin experiment
Lab 2	Virtual quantum lab- Laser beam experiment

Lab 3	Mach-Zehnder interferometer
Lab 4	Introduction to quantum simulator
Lab 5	Construction of Bell states
Lab 6	Quantum simulations of classical circuits
Lab 7	Binary Oracles
Lab 8	Phase Oracles
Week 9	**Mid-Semester Break**
Lab 10	Deutsch problem
Lab 11	Deutsch-Jozsa problem
Lab 12	Quantum teleportation
Lab 13	Grover's search problem
Lab 14	Application of Grover's search-Quantum cryptanalysis
Lab 15	BB84 algorithm
Lab 16	Open-Ended Lab
Lab 17	Presentation/Demo
Week 18	**End Semester Break**

h. Recommended Reading (including Textbooks and Reference books with dates).

Textbook:

1. Hughes, Ciaran, et al. Quantum computing for the quantum curious. Springer Nature, 2021.
2. Yanofsky, Noson S., and Mirco A. Mannucci. Quantum computing for computer scientists. Cambridge University Press, 2008.
3. Kaye, Phillip, Raymond Laflamme, and Michele Mosca. An introduction to quantum computing. OUP Oxford, 2006.

Reference Material:

1. IBM Quantum platform courses
2. IBM Quantum platform tutorial

i. Assessments

- Quizzes
- Assignments
- MSE

- Project (Optional)
- ESE

Resources

11. No additional requirement

Approval by DBS/FBS

12. Approved by DBS

Miscellaneous

13. N/A

Prepared by Dr Madiha Khalid

Date_____

Checked by _____

Date_____

Approved by _____

Date_____